

Engineering Mathematics Study Notes: Raabe's Test

1. IDENTIFY

The mathematical concept presented on the board is **Raabe's Test**. It is a higher-order convergence test used for infinite series of positive terms. It is primarily executed as a fallback verification when **D'Alembert's Ratio Test** fails because the ratio limit evaluates exactly to 1.

2. THE FORMULA

Let $\sum u_n$ be an infinite series consisting entirely of positive terms. Raabe's Test requires calculating the limit value l using the following expression:

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = l$$

The behavior of the infinite series is evaluated based on the numerical value of l :

1. The series $\sum u_n$ is **convergent** if:

$$l > 1$$

2. The series $\sum u_n$ is **divergent** if:

$$l < 1$$

3. The test **fails** to provide a conclusion if:

$$l = 1$$

3. THE ALGORITHM

This systematic blueprint outlines how to solve convergence problems when basic ratio tests fail.

Step 1: Establish the General Term

Identify the algebraic rule for the n -th term of the given series, designated as u_n .

Step 2: Establish the Successor Term

Obtain the $(n + 1)$ -th term, designated as u_{n+1} , by substituting every instance of n in u_n with $(n + 1)$.

Step 3: Attempt the Base Ratio Test

Set up and compute the limit of the term ratio as n approaches infinity to verify if Raabe's Test is required.

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$$

If this evaluates to 1, the standard test fails, creating a direct dependency on Raabe's Test.

Step 4: Construct the Raabe Ratio

Invert the calculated ratio fraction to get $\frac{u_n}{u_{n+1}}$, subtract 1 from it, and multiply the entire expression by n .

Step 5: Compute the Limit

Evaluate the limit of the new expression as $n \rightarrow \infty$.

$$l = \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$$

Step 6: Apply the Decision Rule

Compare the computed value of l against the reference threshold of 1 to conclude whether the series converges or diverges.

4. THE MECHANICS

Core Operations Explained

- **Term Substitution:** To transition from u_n to u_{n+1} , wrap $(n + 1)$ in parentheses wherever n appears. For example, if a term is $2n$, its successor is $2(n + 1) = 2n + 2$. If a term is $2n - 1$, its successor is $2(n + 1) - 1 = 2n + 2 - 1 = 2n + 1$.
- **Fraction Division via Multiplication:** Dividing by a fraction is achieved by multiplying by its reciprocal.

$$\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \times \frac{D}{C}$$

- **Factorial-Style Product Cancellation:** In series containing expanding products of numbers like $1 \cdot 3 \cdot 5 \dots (2n - 1)$, the terms in u_{n+1} contain all factors of u_n plus an additional trailing term. When computing ratios, all common matching front factors cancel out completely, leaving only the individual trailing elements.
- **Infinity Limit Mechanism:** To evaluate limits where $n \rightarrow \infty$, factor out the highest power of n from both the numerator and denominator to isolate expressions of the form $\frac{k}{n}$. As n becomes infinitely large, any term divided by n becomes 0.

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Full Mechanics Walkthrough: Problem 1

Consider the series:

$$\sum \frac{1}{n^2}$$

The general term u_n is defined as:

$$u_n = \frac{1}{n^2}$$

The successor term u_{n+1} is constructed by replacing n with $n + 1$:

$$u_{n+1} = \frac{1}{(n + 1)^2}$$

Evaluate the standard limit ratio of the successor term to the general term:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n^2}}$$

Convert the fraction division into a reciprocal multiplication step:

$$\lim_{n \rightarrow \infty} \frac{1}{(n + 1)^2} \times n^2 = \lim_{n \rightarrow \infty} \frac{n^2}{(n + 1)^2}$$

Factor out n from the base expression within the denominator:

$$\lim_{n \rightarrow \infty} \frac{n^2}{n^2 \left(1 + \frac{1}{n}\right)^2}$$

Cancel out the common n^2 variable shared between numerator and denominator:

$$\lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^2}$$

Substitute the limit variable boundary values into the expression:

$$\frac{1}{\left(1 + \frac{1}{\infty}\right)^2} = \frac{1}{(1 + 0)^2} = 1$$

Because the limit equals 1, D'Alembert's Ratio Test fails. Transition to the Raabe's Test formula:

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$$

Substitute the explicit definitions of u_n and u_{n+1} into the test:

$$\lim_{n \rightarrow \infty} n \left(\frac{\frac{1}{n^2}}{\frac{1}{(n+1)^2}} - 1 \right)$$

Simplify the internal complex fraction inside the parentheses:

$$\lim_{n \rightarrow \infty} n \left(\frac{(n+1)^2}{n^2} - 1 \right)$$

Expand the squared binomial expression located in the numerator:

$$\lim_{n \rightarrow \infty} n \left(\frac{n^2 + 2n + 1}{n^2} - 1 \right)$$

Combine the terms across a unified common denominator:

$$\lim_{n \rightarrow \infty} n \left(\frac{n^2 + 2n + 1 - n^2}{n^2} \right)$$

Simplify the terms inside the numerator by canceling out n^2 :

$$\lim_{n \rightarrow \infty} n \left(\frac{2n + 1}{n^2} \right)$$

Multiply the outer factor n directly into the internal fraction numerator:

$$\lim_{n \rightarrow \infty} \frac{n(2n + 1)}{n^2}$$

Distribute n or cancel one factor of n from the denominator:

$$\lim_{n \rightarrow \infty} \frac{2n + 1}{n}$$

Split the fraction into individual component terms:

$$\lim_{n \rightarrow \infty} \left(2 + \frac{1}{n} \right)$$

Evaluate the limit explicitly as n approaches infinity:

$$2 + 0 = 2$$

Since the final limit value $l = 2$, which satisfies $l > 1$, the series is **convergent**.

Full Mechanics Walkthrough: Problem 2

Consider the series:

$$\frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \dots$$

The general term u_n representing the pattern of the product sequence is:

$$u_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n}$$

The successor term u_{n+1} includes the next sequence terms:

$$u_{n+1} = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots 2n(2n+2)}$$

Set up the standard ratio test limit expression:

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{\frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots 2n(2n+2)}}{\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n}}$$

Convert the division into direct multiplication by using the reciprocal:

$$\lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots 2n(2n+2)} \times \frac{2 \cdot 4 \cdot 6 \dots 2n}{1 \cdot 3 \cdot 5 \dots (2n-1)}$$

Cancel all matching terms appearing simultaneously in numerators and denominators:

$$\lim_{n \rightarrow \infty} \frac{2n+1}{2n+2}$$

Factor out the variable n from both components to prepare for infinity evaluation:

$$\lim_{n \rightarrow \infty} \frac{n \left(2 + \frac{1}{n}\right)}{n \left(2 + \frac{2}{n}\right)}$$

Cancel the isolated factor n from the ratio expression:

$$\lim_{n \rightarrow \infty} \frac{2 + \frac{1}{n}}{2 + \frac{2}{n}}$$

Apply the limit condition values as n goes to infinity:

$$\frac{2 + \frac{1}{\infty}}{2 + \frac{2}{\infty}} = \frac{2+0}{2+0} = 1$$

The ratio test value equals 1, confirming a test failure. Initiate the Raabe's Test process:

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right)$$

Substitute the reciprocal of the simplified ratio term directly into the test:

$$\lim_{n \rightarrow \infty} n \left(\frac{2n+2}{2n+1} - 1 \right)$$

Find a common denominator to unify the elements inside the bracket:

$$\lim_{n \rightarrow \infty} n \left(\frac{(2n+2) - (2n+1)}{2n+1} \right)$$

Simplify the numerator expression by distributing the negative sign:

$$\lim_{n \rightarrow \infty} n \left(\frac{2n+2-2n-1}{2n+1} \right)$$

Combine constants to simplify the remaining expression values:

$$\lim_{n \rightarrow \infty} n \left(\frac{1}{2n+1} \right)$$

Multiply the external factor n into the fractional expression:

$$\lim_{n \rightarrow \infty} \frac{n}{2n + 1}$$

Factor out n from the expression terms located in the denominator:

$$\lim_{n \rightarrow \infty} \frac{n}{n \left(2 + \frac{1}{n} \right)}$$

Cancel out the variable n present across both fields:

$$\lim_{n \rightarrow \infty} \frac{1}{2 + \frac{1}{n}}$$

Apply the final limit condition boundaries to resolve the constant expression:

$$\frac{1}{2 + \frac{1}{\infty}} = \frac{1}{2 + 0} = \frac{1}{2}$$

Compare the result against the decision criterion rule:

$$\frac{1}{2} < 1$$

Since the computed limit value satisfies $l < 1$, the series is definitively **divergent**.